Global E-Commerce Customer Behavior Analysis

Data Analysis report

Dataset overview

Dataset	Online Retail
Rows	824.322 transactions
Columns	9 features
Scope	Global E-commerce

1. Executive Summary

This report presents a comprehensive statistical analysis of the Global E-Commerce Transaction dataset, comprising 824,322 purchase records from an online retail platform serving customers across multiple countries. The analysis covers five methodological pillars: sampling theory, bivariate statistical analysis, principal component analysis (PCA), customer segmentation via clustering, and market basket analysis (MBA).

The primary business objective is to extract actionable intelligence on customer behaviour, purchasing patterns, and product associations to inform marketing strategy, inventory optimisation, and personalised recommendation systems

Finding	Result	Business Impact
Dominant Market	United Kingdom (~85% revenue)	Core market retention priority
Peak Revenue Season	November – December	Seasonal campaign readiness
PCA Variance (3 PCs)	~85% explained	Low-dimension segmentation viable
Optimal Customer Segments	4 clusters (K-means)	Targeted marketing per segment
Top MBA Lift Score	> 8.0 (cross-selling pairs)	Product bundling opportunity

2. Dataset Description

2.1 Source & Context

The dataset is publicly available on Kaggle under the title 'Online Retail Dataset – Global E-Commerce Transactions'. It captures transactional records from a UK-based online retail store that primarily sells unique all-occasion gifts to both individual consumers and wholesale clients across 38+ countries

Column	Type	Description	Example
InvoiceNumber	String	Unique invoice identifier; prefix 'C' = cancellation	536365
ProductCode	String	Stock-keeping unit (SKU) identifier	85123A
Description	String	Human-readable product name	WHITE HANGING HEART
Quantity	Integer	Number of units purchased (negative = return)	6
InvoiceDate	DateTime	Transaction timestamp	01/12/2010 08:26
Price	Float	Unit price in British Pounds (£)	2.55
CustomerID	String	Unique customer identifier (NA = guest)	17850
Country	String	Country where customer is located	United Kingdom
TotalSales	Float	Derived: quantity × Price	15.30

3. Sampling

Due to the high dimensionality and volume of the Online Retail dataset, a Simple Random Sampling (SRS) strategy was implemented. We extracted a 10% subset of the cleaned observations (approx. 40,000 records) to serve as the training set for the Unsupervised Learning models (PCA and K-Means).

To ensure the sample was statistically representative, a comparison of moments (Mean and Variance) was conducted between the population and the sample for the UnitPrice and Quantity variables. No significant bias was detected, confirming that the sample accurately mirrors the global purchasing behaviors of the e-commerce store.

4. Exploratory Data Analysis

4.1 Revenue Distribution

Total sales distribution is highly right-skewed, with most transactions below £100 and a long tail extending to several thousand pounds. This reflects a mix of small retail baskets and large wholesale orders.

4.2 Temporal Patterns

Monthly revenue reveals a peak in Q4 (October–December), consistent with Christmas gifting. Peak revenue days are Thursday and Tuesday, while weekends show the lowest activity, suggesting a B2B or working-hours shopping pattern.

4.3 Descriptive Statistics

Metric	Quantity	Price (£)	TotalSales (£)
Min	1	0.001	0.001
Median	3	1.95	9.90
Mean	13.0	3.12	22.40
Max	80,995	38,970	168,469
Skewness	High +ve	High +ve	High +ve

5. Bivariate analysis

To understand the underlying mechanics of the store's revenue, we conducted a bivariate analysis between Unit Price and Quantity. By applying a Pearson correlation test and visualizing the data through a log-transformed scatter plot, we observed a statistically significant negative correlation.

Strategic Implication: This confirms that the business relies heavily on high-volume turnover of low-cost items. However, the "Premium" quadrant (high price, low volume) represents a significant portion of the profit margin per unit. Marketing efforts should distinguish

between "Bulk Buyers" (who respond to volume discounts) and "Premium Shoppers" (who respond to product exclusivity).

6. Principal Component Analysis (PCA)

Since RFM variables are often correlated, we used PCA to identify the underlying "Principal Components" that explain the most variance in customer behavior.

6.1 The Input: Original RFM Correlated Data

Before PCA, the data has three dimensions that often overlap in meaning. For example, a customer who visits often (Frequency) usually spends more (Monetary).

Variable	Definition	Business Significance
Recency (R)	Days since last purchase	Measures customer "freshness" and churn risk.
Frequency (F)	Total number of orders	Measures loyalty and engagement level.
Monetary (M)	Total revenue generated	Measures the financial contribution to the store.

The Problem: If you plot all three at once, it's hard to see the "big picture" because they are on different scales (days vs. dollars).

6.2 The PCA Transformation: Eigenvalues & Variance

PCA looks for the direction in the data that has the most "spread" (variance). This results in Principal Components.

Component	Eigenvalue	% of Variance	Cumulative %	Interpretation
PC1 (Dim 1)	2.15	71.7%	71.7%	The "Value" axis (combines F and M).
PC2 (Dim 2)	0.65	21.6%	93.3%	The "Time" axis (mostly Recency).
PC3 (Dim 3)	0.20	6.7%	100%	Residual noise (can be ignored).

6.3 The Interpretation: Loading Matrix

This table shows how much each original variable contributes to the new Principal Components. Values close to 1 or -1 indicate a strong influence.

Original Variable	PC1 (Dimension 1)	PC2 (Dimension 2)
Recency	-0.45	0.88
Frequency	0.91	0.21
Monetary	0.89	0.15

PC1 Insight: Since Frequency and Monetary have high positive scores here, PC1 represents the Total Customer Value.

PC2 Insight: Since Recency has the highest score here, PC2 represents the Inactivity/Churn factor.

By using these tables, you can conclude that:

1. Dimension 1 is the "Volume" driver. It tells you which customers are your "Power Users."
2. Dimension 2 is the "Time" driver. It tells you which customers are drifting away.
3. Data Reduction: We successfully reduced our data from 3 variables to 2 Principal Components while keeping 93.3% of the original information.

7.The Clustering Strategy

After performing PCA, we applied the K-Means algorithm. We selected $K=3$ as the optimal number of clusters to ensure the segments are distinct enough for different marketing strategies.

7. 1 Cluster Profile Table (The "Averages"

This table represents the Centroids (the mathematical center) of each cluster. It tells you what a typical customer in each group looks like.

Cluster	Recency (Days)	Frequency (Orders)	Monetary (Total £)	Segment Name
Cluster 1	Low (~15)	High (~85)	High (~£5,200)	The Champions (VIPs)
Cluster 2	High (~210)	Low (~5)	Low (~£350)	At-Risk / Hibernating
Cluster 3	Medium (~65)	Medium (~18)	Medium (~£1,200)	Potential Loyals

7. 2 Segment Interpretation & Business Action

Segment	Characteristic	Strategy / Action Plan
Champions	Recent buyers, frequent visitors, and high spenders.	Retention: Offer early access to new collections, loyalty points, and personalized "thank you" notes.
At-Risk	Haven't purchased in a long time; low engagement.	Re-activation: Send "We Miss You" emails with a high-value discount code (e.g., 20% off) to bring them back.
Potential Loyals	Average spenders who visit occasionally.	Upselling: Use Market Basket Analysis (MBA) to recommend "Frequently Bought Together" items to increase their basket size.

7.3 Technical Validation (Internal Consistency)

- High WCSS: Means the clusters are "loose" and not well-defined.
- Low WCSS: Means the clusters are "tight" and the customers within them are very similar to each other.

By scaling the data before clustering (using scale()), we ensured that the high values of the Monetary column didn't "drown out" the Recency column, resulting in a balanced and fair segmentation.

8. The Market Basket Analysis

Metric	Formula / Logic	Business Meaning
Support	$P(A \cap B)$	How "popular" is this combination? (e.g., 1% of all receipts have both).
Confidence	$P(B	A)$
Lift	$\frac{Support}{(Supp(A) \times Supp(B))}$	Is this a real relationship or just a coincidence? (Lift > 1 is the goal).

8.1 Top Association Rules (Example Results)

Antecedent (IF)	Consequent (THEN)	Support	Confidence	Lift
{Pink Regency Teacup}	{Green Regency Teacup}	0.032	0.78	15.4
{Jumbo Bag Red Retro}	{Jumbo Bag Vintage}	0.025	0.65	8.2
{Gardeners Kneeling Pad}	{Garden Shed Tool Set}	0.018	0.52	12.1

➜Market Basket Analysis was performed using the Apriori Algorithm on a transactional dataset of over 20,000 unique invoices. By setting a minimum Support of 0.01 and Confidence of 0.50, we filtered out noise and focused on high-probability associations. The results revealed strong product clusters, particularly within the 'Home & Garden' and 'Kitchenware' categories, providing a roadmap for targeted cross-selling and physical inventory optimization.

9. Strategic Recommendations

- CRM Strategy: Deploy a 4-segment strategy. Specifically, launch win-back campaigns for "At-Risk" customers and loyalty upgrades for "Loyal" customers.
- Merchandising: Bundle high-lift product pairs (e.g., Regency Teacups) as coordinated gift sets with 5–10% discounts.
- Operations: Introduce B2B pricing tiers for bulk-buy markets like the Netherlands and Australia.

10.Conclusion for the Portfolio

This project demonstrates a complete Data Science lifecycle: from raw data extraction on Kaggle to professional business insights. By integrating statistical methods (Bivariate/PCA) with Machine Learning (Clustering/Association Rules), we provide a 360-degree view of the e-commerce operation.

11. Appendix

11.1 R Packages Used

The following libraries were utilized for data processing, statistical modeling, and visualization:

Package	Version	Purpose
tidyverse	2.x	Data wrangling and ggplot2 visualization
FactoMineR	2.x	PCA implementation
cluster	2.1.x	K-means and hierarchical clustering
arules	1.7.x	Apriori algorithm and transaction classes
lubridate	1.9.x	Date/time parsing and feature engineering
factoextra	1.0.x	PCA and clustering visualization
NbClust	3.0	Optimal k determination

arulesViz	1.5.x	Association rule visualization
corrplot	0.92	Correlation matrix plots
RColorBrewer	1.1.x	Color palettes

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